

Computing the BIC Score

$X_v = (X_1, \dots, X_d)$ is generated from a linear Gaussian SCM; i.e. $\forall j \in [1, \dots, d]$

$$X_j = \sum_{\substack{h \in \text{Pa}(j) \\ \text{in } \mathcal{G}}} B_{jh} X_h + \varepsilon_j \quad \varepsilon_j \sim \mathcal{N}(0, \sigma_j^2)$$

Weights:

$$B \in \mathbb{R}^{d \times d} : B_{ij} \neq 0 \iff j \rightarrow i \in \mathcal{G}$$

Note that B is constrained by $\mathcal{G}$!

Variances:

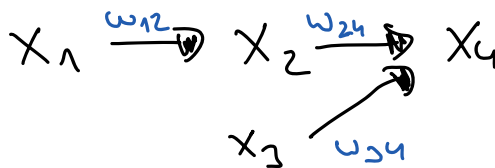
$$\Omega \in \mathbb{R}^{d \times d}$$

$$\Omega = \begin{pmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \ddots & & \\ \vdots & & \ddots & \\ 0 & \dots & 0 & \sigma_d^2 \end{pmatrix}$$

Model Parameters

$$\theta_{\mathcal{G}} = (B, \Omega)$$

Example



$$X_1 := \varepsilon_1$$

$$X_2 := w_{12} X_1 + \varepsilon_2$$

$$X_3 := \varepsilon_3$$

$$X_4 := w_{24} X_2 + w_{34} X_3 + \varepsilon_4$$

$$X = BX + \varepsilon \text{ when } B = \begin{pmatrix} 0 & 0 & 0 & 0 \\ w_{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & w_{24} & w_{34} & 0 \end{pmatrix}, \quad \Omega = \begin{pmatrix} \sigma_1^2 & & & \\ & \ddots & & \\ & & \sigma_4^2 & \end{pmatrix} \Rightarrow \forall i \neq j: \varepsilon_i \perp \varepsilon_j$$

$$\varepsilon = (\varepsilon_1, \dots, \varepsilon_4) \sim \mathcal{N}(0, \Omega)$$

Given $\mathcal{G} = (V, E)$, parameters $\theta_{\mathcal{G}} = (B, \Omega)$,

$$p(X_{V_j} | \theta_{\mathcal{G}}, \mathcal{G}) = \prod_{j \in [d]} \underbrace{p(X_j | X_{\text{Pa}(j)}; \theta_{\mathcal{G}})}_{\sim \mathcal{N}\left(\sum_{k \in \text{Pa}(j)} B_{jk} X_k, \sigma_j^2\right)}$$

Example: $p(X_4 | X_2, X_3; \theta_{\mathcal{G}}) \sim \mathcal{N}(w_{24}X_2 + w_{34}X_3, \sigma_4^2)$

BIC Score

Consider the data $\mathcal{D}$ fixed:

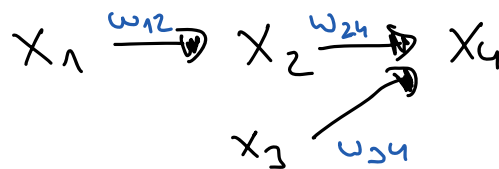
$$\hat{\theta}_{\text{MLE}} = \underset{\theta_{\mathcal{G}}}{\operatorname{argmax}} \sum_{i=1}^n \sum_{j=1}^d \log p(x_j^{(i)} | x_{\text{Pa}(j)}^{(i)}; \theta_{\mathcal{G}})$$

Parameters for each variable j : $B_{jk}: k \in \text{Pa}(j), \sigma_j^2$
(find by OLS regression on parents)

$$S(D; \mathcal{G}) = \underbrace{2 \sum_{i=1}^n \sum_{j=1}^d \log p(x_j^{(i)} | x_{\text{Pa}(j)}^{(i)}; \hat{\theta}_{MLE})}_{2 * \text{log-likelihood of MLE}} - \log n (\underbrace{\# \text{parameters}}_{= d + \# \text{edges}})$$

$\uparrow$
 d variances
 $\uparrow$
 $\# \text{edges}$

Example



$$X_1 := \varepsilon_1$$

$$X_2 := w_{12} X_1 + \varepsilon_2$$

$$X_3 := \varepsilon_3$$

$$X_4 := w_{24} X_2 + w_{34} X_3 + \varepsilon_4$$

Create $\mathcal{G}' = \mathcal{G}$ with an additional edge $1 \rightarrow 4$.

By the local Markov property $X_4 \perp\!\!\!\perp X_1 \mid [X_2, X_3]$

$\Rightarrow$ The log-likelihood w.r.t. $\mathcal{G}' \leq \mathcal{G}$,

however, $\# \text{parameters } \mathcal{G}' > \# \text{parameters } \mathcal{G}$

$\Rightarrow S(\mathcal{G}', D) < S(\mathcal{G}, D)$